

Saturation Temperature and Pressure Lab

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Synopsis

The relationship between the saturation temperature and absolute pressure of water was investigated using an Armfield TH3 saturation-pressure apparatus. Nine equilibrium-stage readings were recorded between 50.00 kPa and 450.00 kPa gauge pressure. Gauge pressures were converted to absolute pressure using the measured atmospheric pressure of 101.65 kPa, and PT100 resistance readings were converted to temperature using the laboratory calibration. The initial comparison graph used the experimental points together with the pressure–temperature values stated directly in the saturated-water pressure table. Linear interpolation was used later only to find a theoretical temperature at each experimental absolute pressure for the point-by-point error calculations.

Both datasets followed a smooth power relationship of the form $T = AP^n$. The experimental fit was $T = 37.633P^{0.2146}$, compared with $T = 31.019P^{0.2557}$ for the direct steam-table data. Experimental temperatures were below the theoretical values at every point. Signed error changed from -1.21% at 151.65 kPa absolute to -6.47% at 551.65 kPa absolute, with a mean signed error of -4.32% . The growing negative deviation identifies a systematic effect rather than random scatter, most plausibly incomplete thermal equilibrium, heat loss and probe lag, residual non-condensable air, or uncertainty in the two-point resistance calibration.

1 Aim

To measure the relationship between the saturation temperature and absolute pressure of water, model that relationship with a power law, and assess the accuracy of the apparatus by comparison with published saturated-water data.

2 Theory

At liquid–vapour equilibrium, water and saturated steam coexist at one saturation temperature for a given absolute pressure. Increasing pressure raises the saturation temperature and produces a smooth saturation curve in temperature–pressure space.

The pressure sensor records gauge pressure, so atmospheric pressure must be added:

$$P_{\text{abs}} = P_1 + P_{\text{atm}}, \quad P_{\text{atm}} = 101.65 \text{ kPa}. \quad (1)$$

The PT100 conversion used the two calibration points recorded during the laboratory:

$$\alpha = \frac{R_2 - R_1}{T_2 - T_1} = \frac{138.7 - 109.0}{100 - 15} = 0.35 \, \Omega \, ^\circ\text{C}^{-1}, \quad (2)$$

$$T_{\text{exp}} = T_1 + \frac{R - R_1}{\alpha} = 15 + \frac{R - 109.0}{0.34941}. \quad (3)$$

Over the measured range, the saturation relationship was represented by

$$T_{\text{sat}} = AP_{\text{sat}}^n, \quad (4)$$

where A and n were determined separately for the experimental dataset and the values stated directly in the steam table. Interpolation was not used to construct this comparison dataset.

The theoretical temperature at each experimental pressure was found by linear interpolation:

$$T_{th} = T_a + \frac{P_{abs} - P_a}{P_b - P_a}(T_b - T_a). \quad (5)$$

Signed percentage error was calculated as specified in the laboratory instructions:

$$\text{Error (\%)} = \frac{T_{exp} - T_{th}}{T_{th}} \times 100. \quad (6)$$

3 Equipment

The equipment comprised an Armfield TH3 saturation-pressure apparatus with a 2.40 L boiler, sight glass, closed pipe loop, cartridge heaters, control console, filling and pressure-relief valves, a PT100 resistance thermometer, electronic and Bourdon pressure gauges, a barometer, and de-ionised water.



Figure 1: Armfield TH3 boiler and pipe loop used to produce and measure saturated steam. The photograph records the apparatus, pressure gauge, viewing port, valves and sensor wiring used in the 2026 experiment.

4 Method

1. With the drain and isolating valves closed and console power off, the boiler was filled with de-ionised water to halfway up the sight glass while the filling-point valve remained open.
2. The water was heated to vigorous boiling and steam was vented until the indication stabilised, purging air from the loop. The filling-point valve was then closed to form a constant-volume saturated mixture.

3. Heater input was raised for approximately two minutes and then reduced. Readings were taken only after the PT100 indication stabilised.
4. Gauge pressure and PT100 resistance were recorded at nine stages from 50.00 kPa to 450.00 kPa gauge pressure.
5. Gauge pressures were converted to absolute pressure and resistances to experimental temperature. The experimental points and the values stated directly in Table A-5 were plotted on the same axes and fitted separately.
6. For the error table only, theoretical saturation temperatures were interpolated from Table A-5 at each experimental absolute pressure. Temperature errors were then calculated for all nine points.

5 Results

5.1 Atmospheric pressure

The atmospheric pressure measured on the laboratory day was 101.65 kPa. This value was added to every gauge-pressure reading to obtain the absolute pressures reported below.

5.2 Experimental and steam-table saturation curves

Figure 2 compares the experimental measurements with the pressure–temperature pairs stated directly in Table A-5. The steam-table markers are the published values at 150.00 kPa, 175.00 kPa, 200.00 kPa and the subsequent tabulated pressures through 600.00 kPa; they were not interpolated to match the experimental pressures. Each dataset was fitted independently to Equation 4.

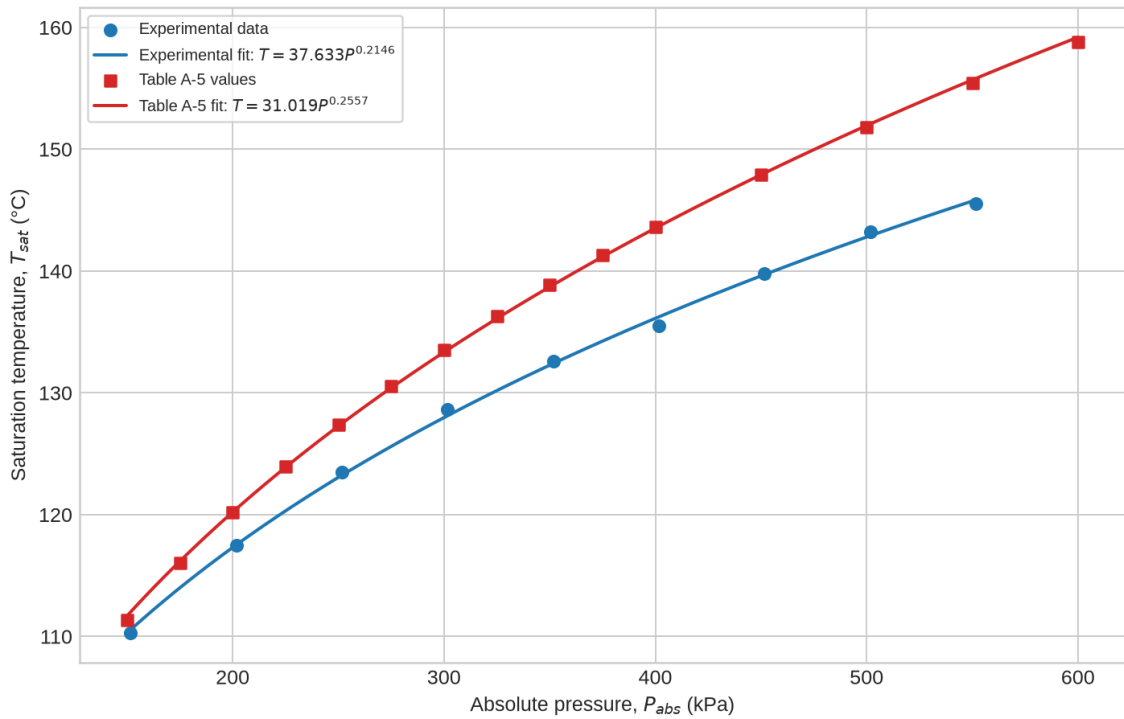


Figure 2: Experimental PT100-derived temperatures and the saturation values stated directly in Table A-5, plotted against absolute pressure with separate power-law trendlines. No interpolated theoretical points are used in this graph.

Table 1: Power-law fit parameters for $T = AP^n$.

Dataset	A	n	R^2
Experimental measurements	37.633	0.2146	0.99895
Direct Table A-5 values	31.019	0.2557	0.99983

5.3 Point-specific interpolation and error calculations

Interpolation was then applied separately to determine T_{th} at each experimental P_{abs} . For the first reading, $P_1 = 50.00 \text{ kPa}$ and $R = 142.30 \Omega$:

$$P_{\text{abs}} = 50.00 + 101.65 = 151.65 \text{ kPa}, \quad (7)$$

$$T_{\text{exp}} = 15 + \frac{142.3 - 109.0}{0.34941} = 110.30^\circ\text{C}. \quad (8)$$

The pressure-table values enclosing 151.65 kPa are 111.35°C at 150.00 kPa and 116.04°C at 175.00 kPa . Therefore,

$$T_{\text{th}} = 111.35 + \frac{151.65 - 150}{175 - 150}(116.04 - 111.35) = 111.66^\circ\text{C}, \quad (9)$$

$$\text{Error} = \frac{110.30 - 111.66}{111.66} \times 100 = -1.21\%. \quad (10)$$

Table 2: Point-specific interpolated temperatures and corresponding errors.

P_1 (kPa gauge)	P_{abs} (kPa)	R (Ω)	T_{exp} ($^\circ\text{C}$)	T_{th} ($^\circ\text{C}$)	Error (%)
50	151.65	142.3	110.30	111.66	-1.21
100	201.65	144.8	117.46	120.46	-2.49
150	251.65	146.9	123.47	127.62	-3.25
200	301.65	148.7	128.62	133.70	-3.80
250	351.65	150.1	132.63	139.02	-4.60
300	401.65	151.1	135.49	143.75	-5.75
350	451.65	152.6	139.78	148.03	-5.57
400	501.65	153.8	143.22	151.95	-5.75
450	551.65	154.6	145.51	155.57	-6.47

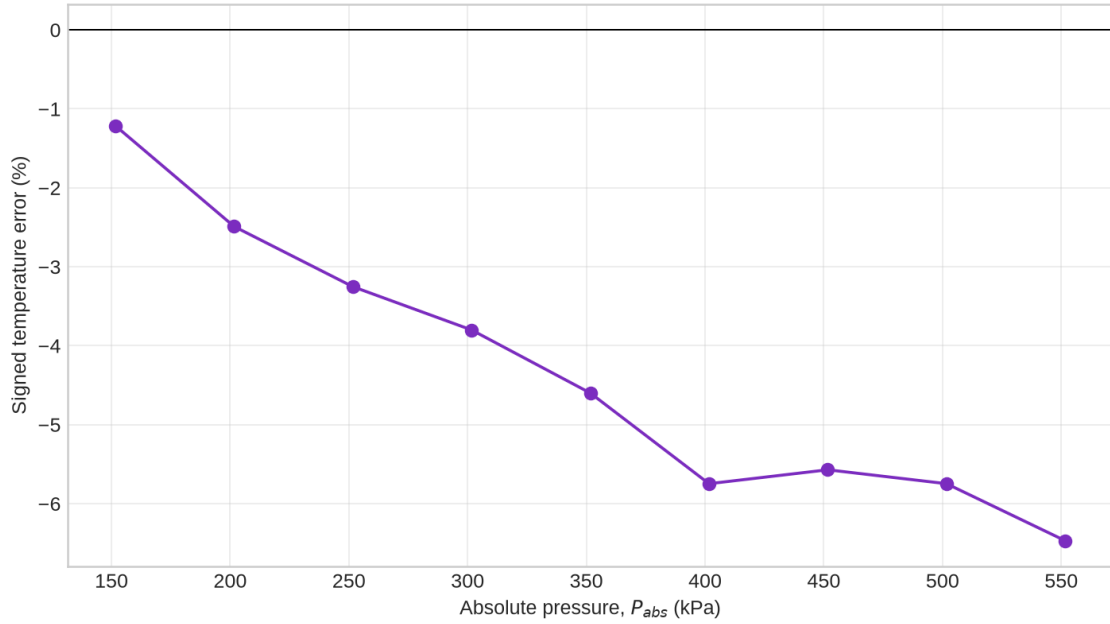


Figure 3: Signed experimental temperature error relative to the point-specific interpolated saturation temperatures.

The mean signed error was -4.32% , and the mean absolute error was 4.32% because every experimental value was below its theoretical value. The maximum absolute error was 6.47% at 551.65 kPa absolute.

6 Discussion

6.1 Comparison of the saturation curves

Both curves increased smoothly with absolute pressure, and both power-law fits had $R^2 > 0.998$, confirming that the measurements captured the expected form of the saturation relationship. However, the experimental curve became progressively lower than the steam-table curve as pressure increased. The mismatch was therefore systematic rather than the result of scattered readings.

6.2 Accuracy and usefulness of the fitted equation

The experimental exponent, $n = 0.2146$, was lower than the direct-table fit of 0.2557 . Consequently, the experimental equation predicts too little temperature rise as pressure increases. It is useful as an empirical description only within the measured range and for this apparatus calibration. It should not replace the steam tables for design calculations or extrapolation. The theoretical fit follows the reference data more closely and is the more reliable compact approximation over this pressure interval.

6.3 Error trend

Error became generally more negative as pressure rose, changing from -1.21% to -6.47% . A small improvement occurred at 451.65 kPa, but it did not alter the overall trend. This pattern indicates a pressure-dependent bias. Reporting only the mean error would conceal this behaviour; Figure 3 therefore provides stronger evidence than a single average.

6.4 Steam temperature, water temperature and thermal lag

Liquid water and steam have the same saturation temperature only when both phases are in thermodynamic equilibrium at the measured pressure. The PT100 measured steam in the upper pipework rather than liquid in the boiler. During heating, energy must conduct through the fluid, metalwork and probe sheath before the sensor reaches the steam temperature. Heat loss from the exposed loop can also make the local probe temperature lower than the bulk saturation temperature. If readings are taken before full stabilisation, both effects produce a negative temperature error, particularly at higher heater input and pressure.

6.5 Likely sources of error

- **Residual air:** air remaining when the filling valve was closed contributes partial pressure. The sensor then measures steam pressure plus air pressure, while saturation temperature depends on steam partial pressure. This produces a measured temperature below the value expected from total pressure.
- **Thermal lag and insufficient stabilisation:** the increasingly negative error is consistent with the PT100 lagging the changing steam temperature.
- **Heat loss and temperature gradients:** the upper loop and sensor fittings lose heat to the room, so the local steam/probe temperature can be below the boiler saturation temperature.
- **Calibration uncertainty:** derived temperatures depend on the two-point linear conversion recorded on the whiteboard. Uncertainty in R_1 , T_1 , R_2 or sensor linearity affects every result systematically.
- **Pressure measurement:** pressure-sensor calibration, gauge resolution and the atmospheric-pressure reading affect every calculated absolute pressure.
- **Interpolation and rounding:** linear interpolation and recorded precision introduce smaller numerical errors, though these are insufficient to explain the observed trend.

The strongest improvements would be to purge the loop for longer, record only after both resistance and pressure remain stable, use the PT100's certified resistance–temperature relation or a traceable calibration, and repeat each pressure point to quantify repeatability.

7 Conclusion

The experiment reproduced the expected increase in water saturation temperature with absolute pressure. The experimental data were smooth and fitted a power law with $R^2 = 0.99895$, but they did not match the reference curve as closely as their smoothness alone might suggest. Experimental temperatures were below the interpolated steam-table values at all nine points, with a mean signed error of -4.32% and a maximum absolute error of 6.47% . The increasing negative error and lower experimental exponent identify a systematic measurement effect, most likely associated with thermal lag, heat loss, residual air, or resistance-calibration uncertainty. Steam-table data remain the appropriate source for accurate saturation values, while the apparatus successfully demonstrates the qualitative pressure–temperature relationship.

References

1. Y. A. Çengel, M. A. Boles, and M. Kanoğlu, *Thermodynamics: An Engineering Approach*, 9th SI ed., McGraw-Hill, 2020, Table A–5.
2. Armfield Ltd., *TH3 Saturation Pressure Apparatus Instruction Manual*, Issue 15.